Ramy Mohamed Sayed

Software Engineer | Backend Developer

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CAREER OBJECTIVE

Backend/Fullstack Software Engineer specialized in **MERN stack** (MongoDB, Express.js, React.js and Node.js), focused on designing and developing scalable **REST APIs**, backend systems and **cloud-native applications**. Looking forward to building reliable, maintainable and high-performance solutions using modern software engineering practices.

EDUCATION

Bachelor of Engineering 2018 – 2023
Helwan University (HU)
Department: Computer Engineering. Overall Grade: **Very Good**

MILITARY SERVICE

Information Systems Reserve Officer, Egyptian Armed Forces 2024 – 2026
Completed

PROJECTS

InTouch
TypeScript, Node.js, Express.js, MongoDB, Redis, Next.js, React Native, Socket.IO
- **Multi-tenant** full stack and Android SaaS communication platform using a layered architecture, **WebSocket,** private R2 uploads, LiveKit-powered **WebRTC** calls, Push notifications and integrated Gemini-powered **AI assistant** "Echo".
- Designed a **production-minded layered architecture** with clear Controller → Service → Repository separation, shared contracts, centralized authorization, MongoDB transactions, **Redis**-backed distributed state, **BullMQ** jobs and **OpenTelemetry/Sentry** observability.
- Built **95 documented REST API endpoints**, **23 typed Socket.IO events**, **588 automated test cases**, shared Zod contracts and 81 files of **technical documentation** , including ADRs, ERDs, OpenAPI specifications, sequence diagrams, and infrastructure decisions.

WadiWays
Node.js, Express, MongoDB, Pug, Stripe
- Egypt-focused travel booking platform with **RESTful APIs**, **Stripe** checkout, **reviews**, content management, and **role-based** user and admin workflows across four MongoDB collections.
- Implemented **centralized exception handling middleware**, **JWT authentication**, **refresh tokens**, role-based authorization (**RBAC**), and **API rate limiting** following **MVC** architecture and **structured logging**.

Munir
Node.js, Express.js, MongoDB, Flutter
- **RESTful** backend API and Flutter mobile application for delivering **unique positive messages without repetition**.
- Rearchitected the system, **migrated data for 30K+ users from in-production Firestore to MongoDB**.
- Integrated the Flutter client with the backend and Integrated **AI tools** to facilitate application workload and deployed the application to the Google **Play Store** .

SKILLS

Backend Development
Node.js, Express.js, REST APIs, MVC Architecture, ASP.NET Core, Entity Framework Core.

Frontend Development
React.js, Next.js, Tailwind CSS, Razor, Pug.

Database
SQL Server, Firebase NoSQL, MongoDB, PostgreSQL.

Cloud & Tools
Git, Android Studio, Postman, Claude Code, GPT Codex, Figma, Atlassian Jira & Confluence, Obsidian, Microsoft Azure, Notion, Docker, GitHub Actions, Stripe Payments, Jest, Playwright.

Programming Languages
C++, Python, Java, JavaScript, HTML, CSS, TypeScript, C#.

Core Concepts & Architectures
Algorithms, Data Structures, Object-Oriented Programming, SOLID Principles, **Problem Solving** , Design Patterns, ACID Databases, Agile methodologies, Authentication & Authorization (JWT, OAuth 2.0, OpenID Connect), Clean Architecture, N-tier Architecture, Layered Architecture, MVC Architecture.